

# Quinn Asena

Data Scientist & Ecologist

Cary Institute of Ecosystem Studies

Forest Futures Lab, Millbrook, NY

## CONTACT

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in LinkedIn

## LAB AFFILIATIONS

Forest Futures Lab

Cary Institute

Williams Paleoecology

UW-Madison

Perry Lab

U. Auckland

Neotoma Paleoecology DB

Leadership Council

## SKILLS & TOOLS

R · Python · C++ · Bash · Slurm/  
PBS · HPC · Git · SQL · Apache  
Arrow

## INTERESTS

Classical piano · Rock climbing  
· Art

## RESEARCH INTERESTS

I am an **ecologist and data scientist** focused on how ecosystems change over time and space, and on building the data infrastructure needed to study them. My current work includes deep learning methods for **wildfire risk mapping**, and process-based **landscape simulation** with CMIP6 climate scenarios on HPC systems to project forest biodiversity change. My background spans palaeoecology, contemporary ecosystem modelling, and the statistical and computational methods that link the two.

## PROFESSIONAL EXPERIENCE

### Data Scientist

*Cary Institute of Ecosystem Studies*

*Millbrook, NY*

*2025 – Present*

- Automated HPC workflows for process-based landscape simulations across boreal North America under multiple CMIP6 climate scenarios.
- Petabyte-scale data management using SQL and Apache Arrow; integration of climate, soil, and elevation data from distributed archives (NASA DAACs, STAC).
- Developed an interactive interface for collaborators to query simulation outputs and generate visualisations.

### Postdoctoral Research Associate

*University of Wisconsin-Madison*

*Madison, WI*

*2022 – 2024*

- NSF-funded project developing state-space modelling methods to analyse palaeoecological records.
- Abrupt change in ecosystems project led by Jack Williams and Anthony Ives.

### Engagement Specialist

*Centre for eResearch - University of Auckland*

*Auckland, NZ*

*2021 – 2022*

- Delivered seminars on data management, FAIR and CARE principles, and reproducible workflows; provided research-computing support for graduate students and research staff (virtual machines, data storage, analytical workflows).
- Technical skills: Linux, Bash, Slurm, HPC, and version control.

### Data Analyst

*Auckland Council*

*Auckland, NZ*

*2021*

- Analysed citizen science bird count data and produced a technical report.

## RESEARCH EXPERIENCE

### Research Assistant

*University of Auckland*

*Auckland, NZ*

*2021 – 2022*

- Conducted a bibliometric analysis on the topic of climate justice.

### Research Assistant

*Stockholm Environment Institute*

*York, UK*

*2016 – 2017*

- Reconstruction of fire regimes in the North York Moors.
- Microscope analysis and core preparation for charcoal and spheroidal carbonaceous particles.

## TEACHING EXPERIENCE

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### PhD Student Mentor

SUNY

Virtual

2025

- One-on-one mentoring in statistics for palaeoecology.

### Graduate Teaching Assistant & Assistant Coordinator

University of Auckland

Auckland, NZ

2018 – 2020

- Assistant coordinator for Discovering Environmental Modelling.
- Guest lecturer on population growth. Developed a [Shiny application](#) as a lecture aid.
- GTA for: Natural and Human Environmental Systems; Discovering Environmental Modelling; Environmental Science and Management.

## EDUCATION

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### PhD, University of Auckland

School of Environment

Auckland, NZ

2017 – 2021

- Explored virtual ecological methods for generating pseudoproxy data to assess statistical inferences under data uncertainty.
- Supervised by George Perry and Janet Wilmschurst.

### MEnv. Environmental Science

University of York

York, UK

2012 – 2016

- First-class integrated master's degree in environmental science. Master's research involved environmental monitoring using sensor networks and robotics.
- Courses included: Ecological Principles; Biogeography; Applied Ecology and Environmental Management; Climate Change: Science, Observation & Impacts; Research Skills and Statistical Methods.

### Geocomputation & Machine Learning for Environmental Applications

Basilicata University

Matera, Italy

2025

- Certified course in machine learning for spatial analyses.

### Certified Carpentries Instructor

The Carpentries

Global (Virtual)

2023

- Certified to teach and host [Carpentries](#) workshops in data science and computational skills.

### Performance Composition & Theory

Royal Academy of Music

London, UK

2005 – 2010

- Study in performance, composition, and musical theory. Multiple competition awards.
- ABRSM Grade 8 Piano Performance.
- ABRSM Grade 5 Music Theory.

## PUBLICATIONS

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- **Asena, Q.**, Williams, J., Johnson, J., Shuman, B., Stefanova, V., Ives, A. (2026). [Statistical analyses of ecological multinomial time series to identify environmental drivers and biotic interactions](#). *Methods in Ecology and Evolution*.
- **Asena, Q.**, Perry, G.L., Wilmschurst, J. (2026). [Information loss in palaeoecological data from process and observer error](#). *Climate of the Past*.

- Lee, F., Bellvé, A.M., Alena, H., **Asena, Q.**, Perry, G.L. (2025). [Using population viability analysis and fossil records to inform the conservation of pateke \(\*Anas chlorotis\*\)](#). *New Zealand Journal of Ecology*.
- **Asena, Q.**, Perry, G.L., Wilmshurst, J. (2024). [Is the past recoverable from the data? Pseudoproxy modelling of uncertainties in palaeoecological data](#). *The Holocene*.
- Parsons, M., **Asena, Q.**, Johnson, D., Nalau, J. (2024). [A Bibliometric Review and Topic Analysis of Climate Justice Literature: Mapping Trends, Voices, and the Way Forward](#). *Climate Risk Management*. Code repository publicly available.
- Courtney-Mustaphi, C.J., Brahney, J., Aquino-López, M.A., Goring, S., Orton, K., Noronha, A., Czaplewski, J., **Asena, Q.**, Paton, S. (2019). [Guidelines for Reporting and Archiving 210Pb Sediment Chronologies to Improve Fidelity and Extend Data Lifecycle](#). *Quaternary Geochronology*.
- Heinemeyer, A., Burn, W.L., **Asena, Q.**, Jones, A., Ashby, M. (2019). [Response to comment on Peatland carbon stocks and burn history](#). *GEO: Geography and Environment*.
- Heinemeyer, A., **Asena, Q.**, Burn, W.L., Jones, A. (2018). [Peatland carbon stocks and burn history: Blanket bog peat core evidence highlights charcoal impacts on peat physical properties and long-term carbon storage](#). *GEO: Geography and Environment*.
- Simpkins, C.E., Lee, F., Powers, B.F., Anderson, S., **Asena, Q.**, Brock, J.M.R., Perry, G.L. (2018). [Population viability analyses in New Zealand: a review](#). *New Zealand Journal of Ecology*.
- Hamilton, R., Brussel, T., **Asena, Q.**, Bruel, R., Marcisz, K., Słowiński, M., Morris, J. (2018). [Assessing the links between resilience, disturbance, and functional traits in paleoecological datasets](#). *Past Global Changes Magazine*, vol. 26(2), 87.

## SOFTWARE & OPEN SCIENCE

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### multinomialTS

*R package*

- [R package](#) for state-space modelling of multinomially distributed ecological time-series data.

### State-Space Methods for Palaeoecological Hypothesis Testing

*Workshop materials*

- [Open workshop materials](#) for state-space modelling of community and palaeoecological time series.

### Authoring Collaborative Research Projects in Quarto

*Workshop materials*

- [Resources for collaborative research workflows](#) using Git, GitHub, and Quarto (ResBaz 2022–2024).

### fisherR

*R package*

- [R package](#) for calculating Fisher's Information on ecological time-series data.

### Population Growth Explorer

*Shiny app*

- [Interactive app](#) exploring population growth equations for undergraduate teaching.

## CONFERENCES, TALKS & WORKSHOPS

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### Methods in Ecology and Evolution

*2026*

*Virtual*

- Talk (recording available soon): [multinomialTS: a state-space approach for modelling community time series](#)
- Materials: [Talk and workshop resources](#)

### ESA 2026 - Ecological Society of America

*2026*

*Salt Lake City, UT*

- **Speaker:** [Assessing climate-driven compositional shifts in boreal forests using iLand](#)

- Neotoma Outreach** 2026  
*Virtual*
- Talk: [A virtual ecological approach to modelling uncertainties in palaeoecology](#)
  - Workshop: [Hands-on workshop for age-depth modelling using Bchron](#)
- ESA 2025 - Ecological Society of America** 2025  
*Baltimore, MD*
- **Speaker:** [Modelling palaeoecological datasets using a state-space approach: two case studies](#)
- AGU - American Geophysical Union** 2025  
*New Orleans, LA*
- Poster: Assessing climate-driven compositional shifts in boreal forests using iLand.
- ESA 2024 - Ecological Society of America** 2024  
*Long Beach, CA*
- **Speaker:** Hypothesis testing of the drivers of long-term ecological change in Michigan.
  - **Workshop host:** [State-space methods for testing hypotheses about environmental drivers and biotic interactions](#)
- INQUA 2023 - International Union for Quaternary Research** 2023  
*Rome, Italy*
- **Lead Convenor:** [Data science and palaeoecology: current intersections and advances](#)
- ESA 2023 - Ecological Society of America** 2023  
*Portland, OR*
- **Speaker:** [Modelling palaeoecological community data: a state-space approach](#)
- ResBaz - Research Bazaar** 2021 – 2025  
*Auckland, NZ; Madison, WI*
- Organiser: ResBaz Aotearoa 2021.
  - Workshop lead: [Quarto for collaboration](#) at ResBaz NZ 2022–2024 and in-person Madison, WI, 2023.
- AGU - American Geophysical Union** 2022  
*Chicago, IL*
- Speaker: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.
- AMQUA - American Quaternary Association** 2022  
*Madison, WI*
- Poster: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.
- BES Palaeo in R - British Ecological Society** 2022  
*Virtual*
- Speaker: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudo-proxies.
- INQUA 2019 - International Union for Quaternary Research** 2019  
*Dublin, Ireland*
- Speaker: Pseudoproxy modelling for assessing statistical inferences.

- Durham University - Invited Speaker**  
*Durham, UK*

- Modelling resilience in ecosystems.

**Biological Heritage Science Challenge**  
*University of Canterbury*

*Christchurch, NZ*  
*2017 – 2019*

- Speaker on resilience in ecosystems.

**International Swiss Climate Summer School**  
*University of Bern*

*Bern, Switzerland*  
*2018*

- Poster presentation and participant. Accredited course (3 ECTS credits).

**EcoRe3: Resistance Recovery and Resilience in Long-term Ecological Systems**  
*PAGES - University of Utah*

*Utah, USA*  
*2018*

- Speaker: modelling resilience in ecosystems.

PEER REVIEW

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|----------------------------------------------|-------------|
| <i>Ecology Letters</i>                       | <i>2026</i> |
| <i>Journal of Open Source Software</i>       | <i>2026</i> |
| <i>Nature Communications</i>                 | <i>2025</i> |
| <i>Environmental Research and Technology</i> | <i>2025</i> |
| <i>Quaternary Geochronology</i>              | <i>2024</i> |
| <i>Methods in Ecology and Evolution</i>      | <i>2022</i> |

SERVICE & LEADERSHIP

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- Early Career Representative**  
*Neotoma Leadership Council*

*Virtual*  
*2024 – Present*

- Early career representative on the Neotoma leadership council. Contributing to database governance, ingestion standards, and community user support for a global palaeoecological data resource.